

Ramusthi Vuhwario

Section. 1 - Students

1. SELECT DISTINCT (department)
FROM Students;

Outcome =

department
IT
Finance
HR

2. SELECT department
~~AVG~~ AVG(age)
AS avg-age
FROM Students
GROUP BY departments;

outcome =

department	avg-age
IT	21,5
Finance	22
HR	21,5

3. SELECT department,
 COUNT(*)
 AS student-count
 FROM students
 GROUP BY department
 HAVING COUNT(*) > 1;

outcome =

department	student-count
IT	2
HR	2

4. SELECT student-id,
 name,
 age
 FROM students
 WHERE age BETWEEN 21 AND 23,

outcome =

student-id	name	age
2	Bob	22
3	Charlie	21
4	Alice	23
5	Eve	22

5. SELECT Student-id,
name,
age,
Department
FROM Students
WHERE department IN ('IT', 'HR')
AND age > 21;

outcome =

Student-id	name	age	Department
4	Diana	23	IT
5	Eve	22	HR

Section 2

6. SELECT department,
sum(credits) AS total_credits
FROM courses
GROUP BY department
HAVING sum(credits) > 5;

outcome =

department	total_credits
IT	11

7. **SELECT** course-id,
course-name,
department,
credits
FROM courses
WHERE credits < 4;

~~outcome =~~

course-id	course-name	department	credits
101	SQL basics	IT	3
102	Python	IT	3
105	Statistics	HR	3

outcome =

course-id	course-name	department	credits
101	SQL basics	IT	3
105	Statistics	HR	3
104	Excel	Finance	2

8. **SELECT** course-id,
course-name,
credits
FROM courses
ORDER BY credits **DESC**
LIMIT 3,

outcome =

course-id	course name	credits
103	Data Science	4
102	Python	4
101	SQL Basics	3

Section 3

9. SELECT MAX(grade) AS max-grade,
MIN(grade) AS min-grade,
AVG(grade) AS avg-grade,
FROM enrollments

outcome =

max-grade	min-grade	avg-grade
90	78	84,6

10. SELECT course_id,
COUNT AS enrollment-count
FROM enrollments
Group BY course_id;

outcome =

course-id	enrollment-count
101	1
102	1
103	1
104	1
105	1

Section 4

11. ~~SELECT department~~

11. SELECT department
sum(salary) AS total-salary,
sum(bonus) AS total-bonus.
FROM salaries
GROUP BY department,

outcome =

department	total-salary	total-bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. SELECT department,
AVG(salary) AS avg-salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;

~~outcome~~ =

department	avg-salary
IT	61000
Finance	70000

13. SELECT employee-id,
 name,
 salary,
 bonus,
 (salary + bonus) AS total-compensation
 FROM salaries
 WHERE (salary + bonus) > 60000;

outcome 2

employee-id	name	salary	bonus	total-compensation
1	Tom	60000	5000	65000
3	Spoke	70000	6000	76000
4	Tyre	62000	5500	67500

10. Section 5

14. SELECT department,
 sum(budget) AS total-budget,
 avg(budget) AS avg-budget
 FROM projects
 Group By department
 HAVING avg(budget) > 70000;

outcome 2

department	total-budget	avg-budget
IT	270000	135000
finance	80000	80000

13. ~~SELECT project-id,
project-name,
department,
budget
FROM projects
WHERE budget BETWEEN 50000
AND 120000
AND department~~

15. SELECT project-id,
project-name,
department,
budget
FROM projects
WHERE budget BETWEEN 50000
AND 120000
AND NOT department = 'Marketing';

outcomes

z	project-id	project-name	department	budget
	1	AI APP	IT	120000
	2	Payroll system	Finance	80000
	5	HR Portal	HR	50000